

LNPTM THERMOCOMPTM COMPOUND D551RC

DESCRIPTION

LNP THERMOCOMP D551RC is a compound based on Polycarbonate resin containing Glass Fiber, Flame Retardant. Added features of this material include: High modulus, good flatness, good impact performance, Non-Brominated & Non-Chlorinated Flame Retardant, PCR content up to 30%.

TYPICAL PROPERTY VALUES

Revision 20200521

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
MECHANICAL			
Tensile Strain, brk, Type I, 5 mm/min	2	%	ASTM D 638
Tensile Modulus, 5 mm/min	16090	MPa	ASTM D 638
Flexural Modulus, 1.3 mm/min, 50 mm span	13420	MPa	ASTM D 790
Tensile Stress, break, 5 mm/min	147	MPa	ISO 527
Tensile Strain, break, 5 mm/min	1.35	%	ISO 527
Tensile Modulus, 1 mm/min	15270	MPa	ISO 527
IMPACT			
Izod Impact, unnotched, 23°C	423	J/m	ASTM D 4812
Izod Impact, notched, 23°C	119	J/m	ASTM D 256
Charpy 23°C, Unnotch Edgew 80*10*4 sp=62mm	31	kJ/m ²	ISO 179/1eU
Charpy 23°C, V-notch Edgew 80*10*4 sp=62mm	12	kJ/m ²	ISO 179/1eA
THERMAL			
HDT, 1.82 MPa, 3.2mm, unannealed	105	°C	ASTM D 648
CTE, -40°C to 40°C, flow	1.31E-05	1/°C	ASTM E 831
CTE, -40°C to 40°C, xflow	4.24E-05	1/°C	ASTM E 831
Relative Temp Index, Elec ⁽¹⁾	80	°C	UL 746B
Relative Temp Index, Mech w/impact ⁽¹⁾	80	°C	UL 746B
Relative Temp Index, Mech w/o impact ⁽¹⁾	80	°C	UL 746B
PHYSICAL			
Density	1.618	g/cm ³	ASTM D 792
Mold Shrinkage, flow, 24 hrs	0.05 – 0.2	%	ASTM D 955
Mold Shrinkage, xflow, 24 hrs	0.05 – 0.2	%	ASTM D 955
Melt Volume Rate, MVR at 300°C/2.16 kg	20	cm ³ /10 min	ASTM D 1238
Melt Volume Rate, MVR at 300°C/5.0 kg	34.98	cm ³ /10 min	ASTM D 1238
FLAME CHARACTERISTICS ⁽¹⁾			
UL Yellow Card Link	E207780-102777817	-	-
UL Recognized, 94V-0 Flame Class Rating	≥1	mm	UL 94
INJECTION MOLDING			
Drying Temperature	110	°C	
Drying Time	3 – 6	hrs	
Maximum Moisture Content	0.02	%	
Melt Temperature	285 – 310	°C	
Nozzle Temperature	285 – 305	°C	
Front - Zone 3 Temperature	280 – 300	°C	

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
Middle - Zone 2 Temperature	270 – 290	°C	
Rear - Zone 1 Temperature	260 – 280	°C	
Mold Temperature	80 – 110	°C	
Back Pressure	0.1 – 0.3	MPa	
Screw Speed	50 – 90	rpm	

(1) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.

DISCLAIMER

Any sale by SABIC, its subsidiaries and affiliates (each a "seller"), is made exclusively under seller's standard conditions of sale (available upon request) unless agreed otherwise in writing and signed on behalf of the seller. While the information contained herein is given in good faith, SELLER MAKES NO WARRANTY, EXPRESS OR IMPLIED, INCLUDING MERCHANTABILITY AND NONINFRINGEMENT OF INTELLECTUAL PROPERTY, NOR ASSUMES ANY LIABILITY, DIRECT OR INDIRECT, WITH RESPECT TO THE PERFORMANCE, SUITABILITY OR FITNESS FOR INTENDED USE OR PURPOSE OF THESE PRODUCTS IN ANY APPLICATION. Each customer must determine the suitability of seller materials for the customer's particular use through appropriate testing and analysis. No statement by seller concerning a possible use of any product, service or design is intended, or should be construed, to grant any license under any patent or other intellectual property right.